

STPM PHYSICS • SEMESTER 2 • CHAPTER 13

Capacitors

13.1 Capacitance

DEFINITIONS & LAWS

- **Capacitance** of a capacitor is the ratio of the magnitude of the charge on either plate to the potential difference between the plates ($C = \frac{Q}{V}$).
- **One farad** is the capacitance of a capacitor that stores a charge of one coulomb when the potential difference across it is one volt ($1 \text{ F} = 1 \text{ C V}^{-1}$).
- The capacitance depends only on the geometry of the capacitor (area, separation, shape) and the dielectric between the plates; it does not depend on the charge stored or the potential difference applied.
- When a capacitor is connected to a battery the potential difference V is fixed; when it is isolated (disconnected) the charge Q remains constant.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Capacitance	$C = \frac{Q}{V}$	C = capacitance (F); Q = charge on one plate (C); V = potential difference across the plates (V)
Capacitance of an isolated sphere	$C = 4\pi\epsilon_0 R$	Isolated conducting sphere; R = radius (m); ϵ_0 = permittivity of free space (F m^{-1})
Parallel plate capacitor (air)	$C = \frac{\epsilon_0 A}{d}$	A = area of one plate (m^2); d = plate separation (m)
Parallel plate capacitor (dielectric)	$C = \epsilon_0 \epsilon_r \frac{A}{d}$	ϵ_r = relative permittivity of the dielectric (no unit)
Unit of capacitance	$\text{F} = \text{C V}^{-1} = \text{A}^2 \text{ s}^4 \text{ kg}^{-1} \text{ m}^{-2}$	Typical values are μF (10^{-6} F), nF (10^{-9} F) and pF (10^{-12} F)

- On a graph of Q against V (straight line through the origin) the gradient is C .

HOW TO ANSWER (MPM STYLE)

- State $C = \frac{Q}{V}$, substitute with units converted (μF , μC , mV) and give the answer in F with the suitable prefix.
- For a change question, state first which quantity is constant: V constant (still connected to battery) or Q constant (isolated), then use $Q = CV$ or $V = \frac{Q}{C}$.
- For a definition write "ratio of charge on one plate to the potential difference across the plates", and do not omit the word ratio.
- For a graph question, label the axes and state that the gradient of the Q - V graph equals the capacitance.

SECTION A TRAPS

Trap: Wrong dependence: options stating that C is proportional to Q or to V ; C is constant for a given capacitor.

Trap: Constant quantity: with the battery disconnected, doubling C halves V ; with the battery connected, doubling C doubles Q .

Trap: Sphere sharing: two identical spheres sharing charge keep the same capacitance; $C = 4\pi\epsilon_0 R$ depends on R only.

Trap: Unit slip: a distractor writes the farad with A instead of A^2 , or with s^2 instead of s^4 ; check using $F = C V^{-1}$.

TRICKS

Trick: $Q = CV$ is a direct proportion: if V rises by $\frac{20}{12}$, Q rises by the same factor, and C need not be found.

Trick: Inverse relation: for the same charge a larger V means a smaller capacitance, since $C = \frac{Q}{V}$.

13.2 Parallel Plate Capacitors

DEFINITIONS & LAWS

- For a parallel plate capacitor the capacitance is directly proportional to the area of the plates and inversely proportional to their separation.
- The electric field between the plates is uniform, $E = \frac{V}{d} = \frac{\sigma}{\epsilon_0}$, directed from the positive plate to the negative plate; the edge effects are neglected when d is much smaller than the plate dimensions.
- If the battery stays connected V is constant, so $Q = CV$ changes when d or A changes; if the capacitor is isolated Q is constant.
- A slab of dielectric that partly fills the gap in layers (parallel to the plates) forms capacitors in series; a slab covering part of the area forms capacitors in parallel.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Parallel plate capacitance (air)	$C = \frac{\epsilon_0 A}{d}$	C = capacitance (F); A = area of one plate (m^2); d = plate separation (m); $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
Capacitance with dielectric	$C = \epsilon_0 \epsilon_r \frac{A}{d}$	Dielectric completely filling the gap; ϵ_r = relative permittivity (no unit)
Field between the plates	$E = \frac{V}{d} = \frac{Q}{\epsilon_0 A}$	E = electric field strength ($V m^{-1}$); V = potential difference (V); Q = charge (C)

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Layers in series	$\frac{1}{C} = \frac{d_1}{\epsilon_0 A} + \frac{d_2}{\epsilon_0 \epsilon_r A}$	d_1, d_2 = layer thicknesses (m); ϵ_r = relative permittivity of the second layer
Stored energy	$U = \frac{1}{2} CV^2$	U = energy stored (J); C = capacitance (F); V = potential difference (V)

HOW TO ANSWER (MPM STYLE)

- Write $C = \frac{\epsilon_0 A}{d}$, convert cm^2 to m^2 ($\times 10^{-4}$) and mm to m ($\times 10^{-3}$) before substituting, then answer in F or pF with 3 s.f.
- For change questions use $C \propto \frac{A}{d}$, state what is constant (V or Q) and then use $Q = CV$ or $V = \frac{Q}{C}$.
- For a layered capacitor find each layer's capacitance ($C = \epsilon_0 \epsilon_r \frac{A}{d_{\text{layer}}}$) and combine in series.
- For field strength use $E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$, or $E = \frac{V}{d}$, and state the units V m^{-1} .

SECTION A TRAPS

Trap: Inverted dependence: C increases when d decreases or A increases; options with C increased by increasing d or removing the dielectric are wrong.

Trap: Battery connected: when d increases, C decreases and $Q = CV$ decreases; when d doubles Q is halved, not doubled.

Trap: Area unit: $6.0 \text{ cm}^2 = 6.0 \times 10^{-4} \text{ m}^2$ (not $6.0 \times 10^{-2} \text{ m}^2$); a missing conversion gives an answer wrong by a factor of 10^2 or 10^3 .

Trap: Combined change: halving d and doubling A multiplies C by 4, not 2.

TRICKS

Trick: Ratio method: $\frac{C'}{C} = \frac{A'}{A} \frac{d}{d'}$; the constants ϵ_0 cancel.

Trick: Layered gap: each layer of thickness $\frac{d}{2}$ has twice the air capacitance, so in series the value is quickly found as $\frac{2C_0 \times 6C_0}{2C_0 + 6C_0} = 1.5C_0$ for $\epsilon_r = 3$.

13.3 Dielectrics

DEFINITIONS & LAWS

- A **dielectric** is an insulating material which becomes polarised in an electric field.

- **Relative permittivity (dielectric constant)** ϵ_r is the ratio of the capacitance of a capacitor with the dielectric between the plates to its capacitance when there is a vacuum between them ($\epsilon_r = \frac{C}{C_0} = \frac{\epsilon}{\epsilon_0}$).
- **Polarisation:** in the external field the molecules of the dielectric become aligned (polar molecules) or stretched (non-polar molecules), so that opposite induced charges appear on the faces of the dielectric.
- The induced charges produce an electric field opposite in direction to the applied field, so the net field in the dielectric is reduced ($E = \frac{E_0}{\epsilon_r}$), the potential difference between the plates falls for the same charge and so the capacitance increases.
- **Dielectric breakdown** occurs when the field exceeds the dielectric strength of the material and the insulator begins to conduct.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Capacitance with dielectric	$C = \epsilon_r C_0 = \epsilon_0 \epsilon_r \frac{A}{d}$	C = capacitance (F); C_0 = capacitance without dielectric (F); ϵ_r = relative permittivity (no unit); A = plate area (m^2); d = plate separation (m)
Isolated capacitor (Q constant): field and p.d.	$E = \frac{E_0}{\epsilon_r}$; $V = \frac{V_0}{\epsilon_r}$	Capacitor isolated at constant Q; E_0 , V_0 = field strength ($V m^{-1}$) and p.d. (V) without dielectric
Connected to battery (V constant)	$Q = \epsilon_r Q_0$; $U = \frac{1}{2} \epsilon_r C_0 V^2 = \epsilon_r U_0$; $E = \frac{V}{d}$ unchanged	Battery connected, V constant; Q_0 = charge without dielectric (C); U_0 = energy without dielectric (J); U = energy (J)
Isolated capacitor: stored energy	$U = \frac{Q^2}{2C} = \frac{U_0}{\epsilon_r}$	Isolated, Q constant; U = energy (J); Q = charge (C); C = capacitance (F)
Maximum voltage	$V_{max} = E_{max} d$	V_{max} = maximum safe voltage (V); E_{max} = breakdown field strength ($V m^{-1}$); d = plate separation (m)

HOW TO ANSWER (MPM STYLE)

- Explain a capacitance increase in three points: the dielectric is polarised, the induced charges produce a field opposing the original field, the net field and the potential difference fall for the same charge, so $C = \frac{Q}{V}$ increases.
- State at the start whether the capacitor is connected to the battery (V constant) or isolated (Q constant); the answer depends on it.
- For numerical work use $C' = \epsilon_r C$, then $Q = C'V$ (connected) or $V' = \frac{Q}{C'}$ (isolated) and give the answer with unit.
- Use the word "partially" for the cancelling effect: the net field is reduced, not zero.

SECTION A TRAPS

Trap: Isolated vs connected: inserting a dielectric into an isolated capacitor reduces V, E and U by ϵ_r ; with the battery connected Q and U increase by ϵ_r and V and E stay the same.

Trap: Factor error: energy changes by ϵ_r , not ϵ_r^2 ; a ratio of 1 : 25 or 25 : 1 for $\epsilon_r = 5.0$ is wrong.

Trap: Complete cancellation: options saying the net field in the dielectric is zero, or that the polarisation field reinforces the external field, are wrong.

Trap: Breakdown: $V_{\max}$ depends on $E_{\max}$ and d , not on ϵ_r ; doubling $E_{\max}$ doubles $V_{\max}$.

TRICKS

Trick: Quick table: connected: V same, E same, $Q \propto \epsilon_r$, $U \propto \epsilon_r$. Isolated: Q same, $V \propto \epsilon_r$, $E \propto \epsilon_r$, $U \propto \epsilon_r$.

Trick: Use $U = \frac{1}{2}QV$: if Q is constant, U follows V ; if V is constant, U follows Q .

13.4 Capacitors in Series and in Parallel

DEFINITIONS & LAWS

- In a **parallel** combination all capacitors have the same potential difference across them, and the total charge is the sum of the charges on the capacitors.
- In a **series** combination every capacitor carries the same charge, and the total potential difference is the sum of the potential differences across the capacitors.
- The equivalent capacitance in parallel is the sum of the capacitances; in series the reciprocal of the equivalent capacitance is the sum of the reciprocals.
- In series the potential difference across a capacitor is inversely proportional to its capacitance, so the smaller capacitor has the larger potential difference.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Capacitors in parallel	$C = C_1 + C_2 + C_3 + \dots$	C = total capacitance (F); C_1, C_2, C_3 = individual capacitances (F)
Capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$	C = total capacitance (F); C_1, C_2, C_3 = individual capacitances (F)
Two capacitors in series	$C = \frac{C_1 C_2}{C_1 + C_2}$	C_1, C_2 = capacitances (F)
Series charge and voltage sharing	$Q = CV$; $V_1 = \frac{VC_2}{C_1 + C_2}$; $V_2 = \frac{VC_1}{C_1 + C_2}$	Q = charge on each capacitor (C); V = total p.d. (V); V_1, V_2 = p.d. across C_1 and C_2 (V)
n identical capacitors	Parallel: nC ; series: $\frac{C}{n}$	n = number of identical capacitors, each of capacitance C (F)

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Total stored energy	$U_{\text{total}} = \frac{1}{2}C_{\text{total}}V^2$	U_{total} = energy of the whole arrangement (J); C_{total} = total capacitance (F); V = p.d. across the arrangement (V)

HOW TO ANSWER (MPM STYLE)

- Reduce the network step by step: combine each series or parallel group, redraw, and finish with a single equivalent capacitance.
- For voltage division use $Q = C_{\text{total}} \times V$ (same for each series capacitor) and then $V_1 = \frac{Q}{C_1}$.
- Write each stage with units (μF) and give the final value to 3 s.f.
- State the rule used at each step: "same charge in series" or "same potential difference in parallel".

SECTION A TRAPS

Trap: Inverted rules: the formulae for capacitors are opposite to those for resistors; a distractor adds the capacitances in series.

Trap: Series bound: the series equivalent must be smaller than the smallest capacitor; $4 \mu\text{F}$ and $6 \mu\text{F}$ in series give $2.4 \mu\text{F}$, whereas $10 \mu\text{F}$ (the sum) or $24 \mu\text{F}$ (the product) is wrong.

Trap: Voltage division: in series the smaller capacitor has the larger share: $3 \mu\text{F}$ and $6 \mu\text{F}$ across 12 V give 8 V and 4 V , not 4 V and 8 V .

Trap: Charge in a mixed network: the charge on the single series capacitor equals the sum of the charges on the two parallel capacitors, not equal to each one.

TRICKS

Trick: Bounds: parallel gives more than the largest value; series gives less than the smallest; use this to eliminate options.

Trick: Identical capacitors C : the possible equivalents are $3C$, $\frac{C}{3}$, $\frac{3C}{2}$ and $\frac{2C}{3}$ only.

13.5 Energy Stored in a Charged Capacitor

DEFINITIONS & LAWS

- The **energy stored in a capacitor** is the work done in charging it, equal to the area under the graph of potential difference against charge.
- **Energy density** of the electric field is the energy stored per unit volume of the field between the plates.
- When a capacitor is charged from a battery through a resistor, the battery supplies energy QV , of which $\frac{1}{2}QV$ is stored in the capacitor and $\frac{1}{2}QV$ is dissipated as heat in the resistor, independent of R .

- When a charged capacitor is connected to an uncharged capacitor the charge is conserved but the total energy decreases; the energy lost is dissipated as heat (and radiation).

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Energy stored in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$	U = energy (J); Q = charge (C); V = potential difference (V); C = capacitance (F)
Energy density	$u = U/(Ad) = \frac{1}{2}\epsilon_0 E^2$	u = energy density (J m^{-3}); A = plate area (m^2); d = plate separation (m); E = field strength (V m^{-1}); ϵ_0 = permittivity of free space (F m^{-1})
Energy density in a dielectric	$u = \frac{1}{2}\epsilon_0 \epsilon_r E^2$	ϵ_r = relative permittivity (no unit)
Work done by the battery	$W = QV = CV^2$	W = work done by the battery (J); Q = charge (C); V = p.d. (V); C = capacitance (F)
Parallel sharing	$V = \frac{Q_{\text{total}}}{C_1 + C_2}$	V = common p.d. after sharing (V); Q_{total} = total charge (C); C_1, C_2 = capacitances (F)

- $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$.

HOW TO ANSWER (MPM STYLE)

- For the energy stored write $U = \frac{1}{2}CV^2$ (choose the form matching the data), substitute with SI units, and give the answer in J with 3 s.f.
- For the V-Q graph, sketch a straight line through the origin and shade the triangle, showing that the area $\frac{1}{2}QV$ equals the energy.
- For energy lost on connecting capacitors find the charge conserved, the final V, the final energy and subtract it from the initial energy.
- Explain a loss as "energy is dissipated as heat in the connecting wires/resistor".

SECTION A TRAPS

Trap: Missing $\frac{1}{2}$: $U = QV$ is the energy supplied by the battery, not the energy stored.

Trap: Wrong power: $U \propto V^2$ (so $3V_0$ gives $9E_0$, not $3E_0$) and $U \propto Q^2$ at fixed C.

Trap: Energy conservation: two equal capacitors, one charged, then connected: the final energy is half the initial, not equal to it.

Trap: Sum of energies: the total energy of two separate capacitors is the sum of each $\frac{1}{2}CV^2$, not $\frac{1}{2}(C_1 + C_2)(V_1 + V_2)^2$.

TRICKS

Trick: Choose the form of U : $\frac{1}{2}CV^2$ at constant V , $\frac{Q^2}{2C}$ at constant Q , and $\frac{1}{2}QV$ if both are known.

Trick: Fixed capacitor: ratio method, $\frac{U'}{U} = \left(\frac{V'}{V}\right)^2$ or $\left(\frac{Q'}{Q}\right)^2$.

13.6 Charging and Discharging of a Capacitor

DEFINITIONS & LAWS

- The **time constant** $\tau = RC$ is the time taken for the charge on a discharging capacitor to fall to $\frac{1}{e}$ (about 37%) of its initial value; for charging it is the time for the charge to reach 63% $\left(1 - \frac{1}{e}\right)$ of its final value.
- During discharge the charge, potential difference and current decay exponentially with time; during charging the current decays exponentially while the charge and potential difference rise exponentially towards their final values.
- At the start of charging the capacitor has no potential difference, so the initial current is $I_0 = \frac{V_0}{R}$; when the capacitor is fully charged the current is zero and the potential difference across it equals the EMF.
- The time constant is increased by increasing R or C ; the final charge depends on C and V_0 only.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Discharging a capacitor	$Q = Q_0 e^{-t/RC}$; $V = V_0 e^{-t/RC}$; $I = I_0 e^{-t/RC}$	Q, V, I = charge (C), p.d. (V), current (A) at time t (s); Q_0, V_0, I_0 = initial values; $I_0 = \frac{V_0}{R}$; R = resistance (Ω); C = capacitance (F)
Charging a capacitor	$Q = Q_0(1 - e^{-t/RC})$; $V = V_0(1 - e^{-t/RC})$; $I = I_0 e^{-t/RC}$	$Q_0 = CV_0$ = final charge (C); V_0 = final p.d. (V); I_0 = initial current (A); R = resistance (Ω); C = capacitance (F); t = time (s)
Time constant	$\tau = RC$	τ = time constant (s); R = resistance (Ω); C = capacitance (F); $e^{-1} = 0.368$, $e^{-2} = 0.135$
Time for charge to halve	$t_{1/2} = RC \ln 2 = 0.693RC$	$t_{1/2}$ = time for the charge to halve (s); R = resistance (Ω); C = capacitance (F)
Logarithmic graph (discharge)	Gradient of $\ln Q$ against $t = -\frac{1}{RC}$	$\ln Q$ against t is a straight line of gradient $-\frac{1}{RC}$ (discharge)

HOW TO ANSWER (MPM STYLE)

- Sketch the graph with labelled axes: Q (or V, I) against t , exponential decay from the initial value that tends to zero, or growth from zero to the final value $Q_0 = CV_0$; mark τ .

- Write the equation, substitute $\frac{t}{RC}$ with RC in seconds ($k\Omega \times \mu F = ms$), evaluate the exponential and give the answer with 3 s.f.
- For a time, rearrange with natural logarithms, e.g. $t = -RC \ln \frac{V}{V_0}$ for discharge or $t = -RC \ln(1 - \frac{V}{V_0})$ for charging.
- State that the exponential never reaches zero in a finite time, and that about 5τ is taken as practically complete.

SECTION A TRAPS

Trap: Meaning of τ : after $t = \tau$ the charge in discharge is $\frac{1}{e}$ (37%) of the initial charge, not zero or half; in charging the capacitor reaches 63%, not 37%.

Trap: Dependence on R: doubling R doubles the time; the initial current $\frac{V_0}{R}$ decreases but the final charge CV_0 is unchanged.

Trap: Two time constants: $\frac{Q}{Q_0} = e^{-2} = 0.135$ (about 14%); a distractor gives 0.37 or 0.74.

Trap: Half-life of the current: I falls to $\frac{I_0}{2}$ at $t = RC \ln 2$ (0.693RC), not $\frac{RC}{2}$ or RC.

TRICKS

Trick: Percentages to recall: charging at $\tau = 63\%$, at $2\tau = 86\%$; discharging at $\tau = 37\%$, at $2\tau = 14\%$.

Trick: Unit check: $k\Omega \times \mu F$ gives ms, $k\Omega \times mF$ gives s, $M\Omega \times \mu F$ gives s.